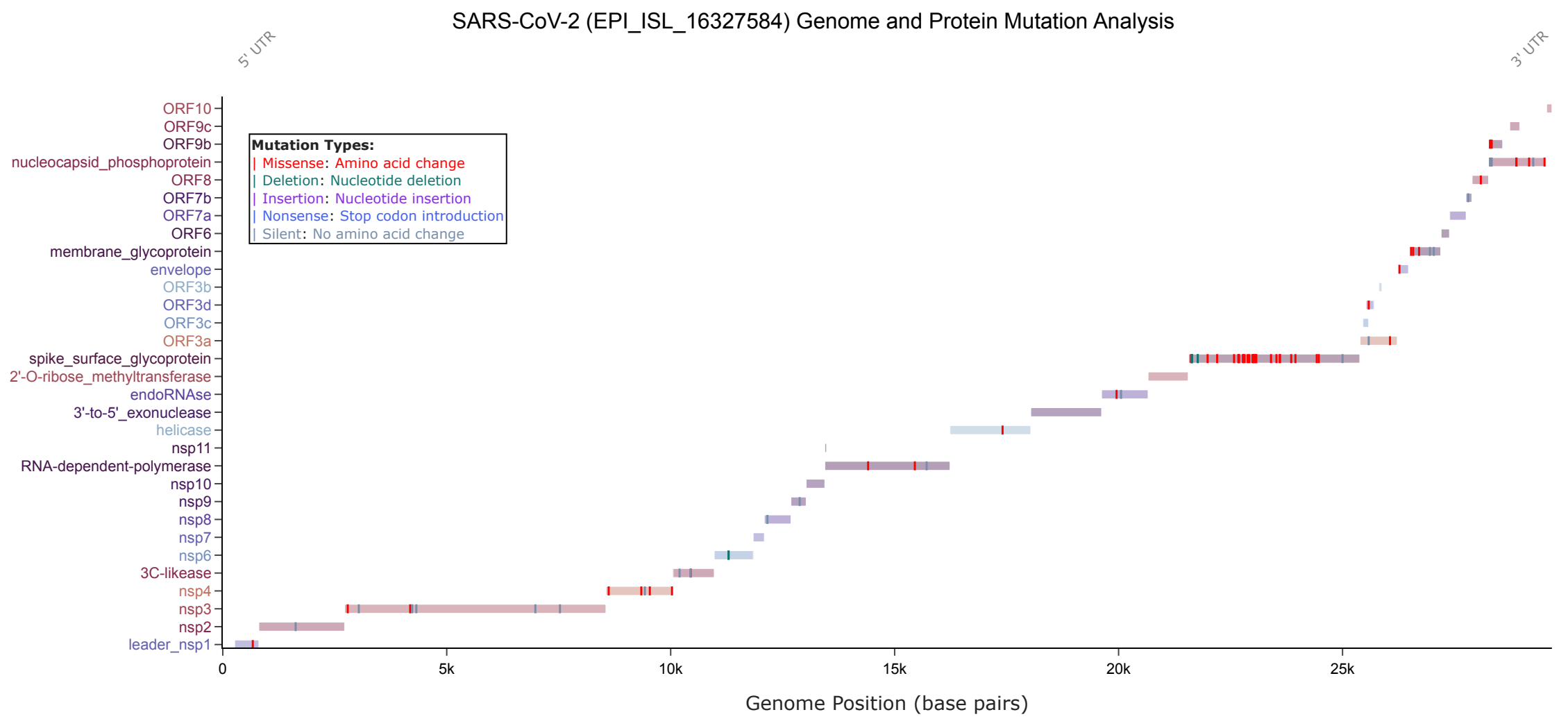
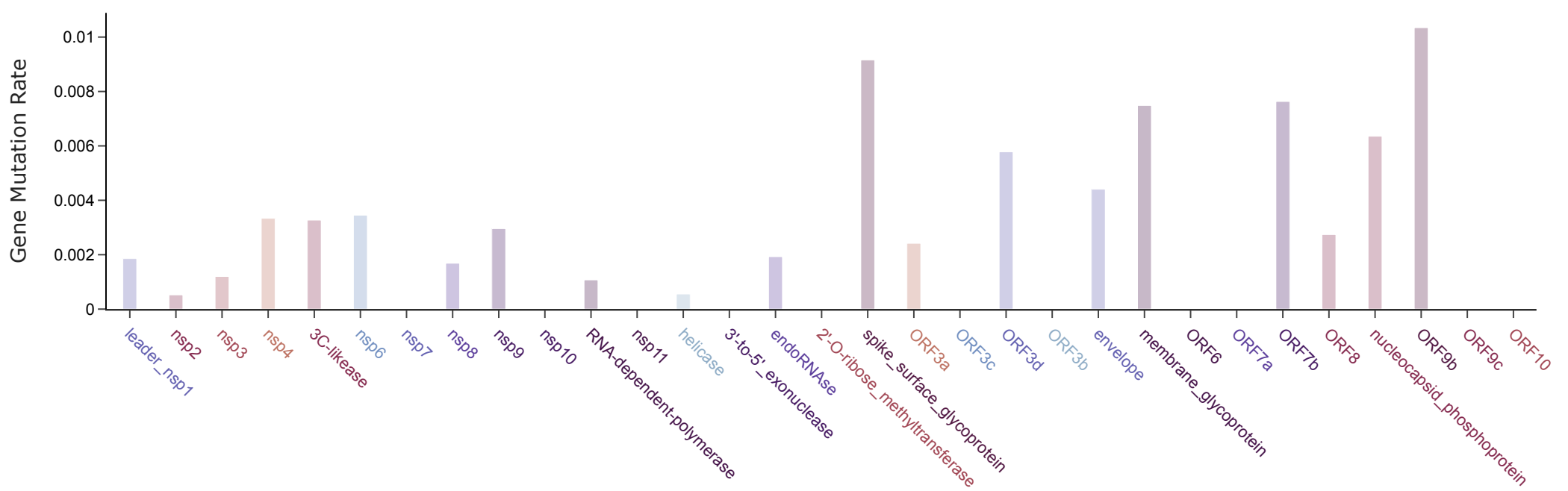
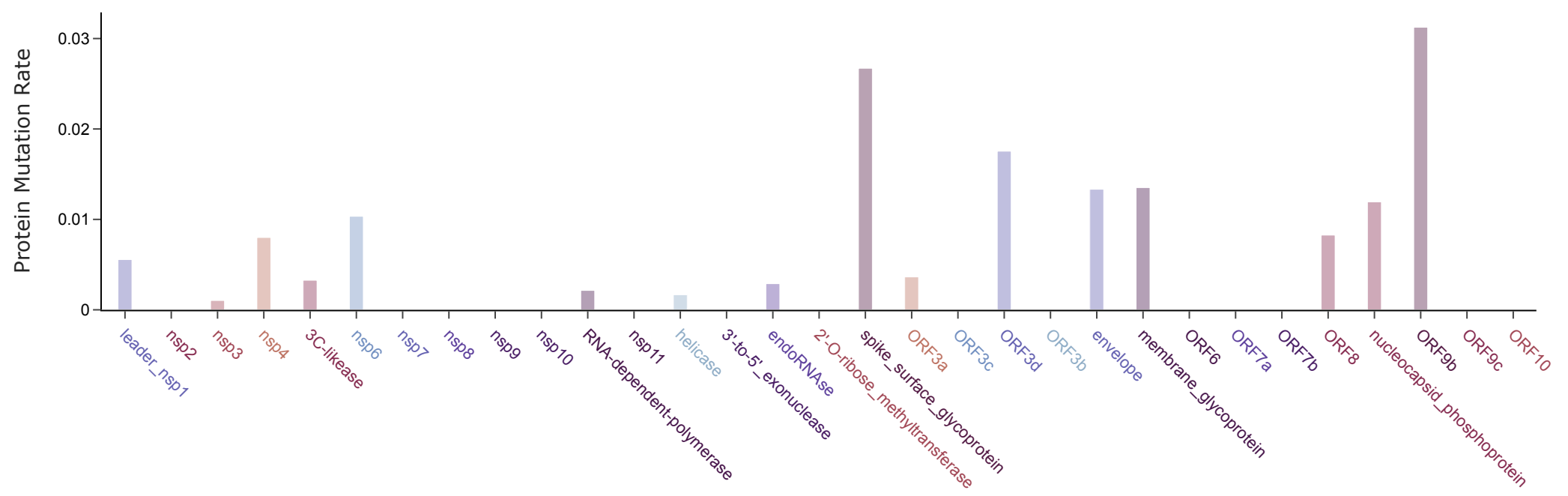

SARS-CoV-2 (EPI_ISL_16327584) Genome and Protein Mutation Analysis
5' UTR
3' UTR
Mutation Types:
| Missense: Amino acid change
| Deletion: Nucleotide deletion
| Insertion: Nucleotide insertion
| Nonsense: Stop codon introduction
| Silent: No amino acid change
ORF10
ORF9c
ORF9b
nucleocapsid_phosphoprotein
ORF8
ORF7b
ORF7a
ORF6
membrane_glycoprotein
envelope
ORF3b
ORF3d
ORF3c
ORF3a
spike_surface_glycoprotein
2'-O-ribose_methyltransferase
endoRNAse
3'-to-5'_exonuclease
helicase
nsp11
RNA-dependent-polymerase
nsp10
nsp9
nsp8
nsp7
nsp6
3C-likease
nsp4
nsp3
nsp2
leader_nsp1
0
5k
10k
15k
20k
25k
Genome Position (base pairs)


Gene Mutation Rate
0.01
0.008
0.006
0.004
0.002
0
leader_nsp1
nsp2
nsp3
nsp4
3C-likease
nsp6
nsp7
nsp8
nsp9
nsp10
RNA-dependent-polymerase
nsp11
helicase
3'-to-5'_exonuclease
endoRNAse
2'-O-ribose_methyltransferase
spike_surface_glycoprotein
ORF3a
ORF3c
ORF3d
ORF3b
envelope
membrane_glycoprotein
ORF6
ORF7a
ORF7b
ORF8
nucleocapsid_phosphoprotein
ORF9b
ORF9c
ORF10


Protein Mutation Rate
0.03
0.02
0.01
0
leader_nsp1
nsp2
nsp3
nsp4
3C-likease
nsp6
nsp7
nsp8
nsp9
nsp10
RNA-dependent-polymerase
nsp11
helicase
3'-to-5'_exonuclease
endoRNAse
2'-O-ribose_methyltransferase
spike_surface_glycoprotein
ORF3a
ORF3c
ORF3d
ORF3b
envelope
membrane_glycoprotein
ORF6
ORF7a
ORF7b
ORF8
nucleocapsid_phosphoprotein
ORF9b
ORF9c
ORF10